

AUTOMATED CHEST X-RAY ANALYSIS REPORT

AI-Assisted Radiology Screening — EfficientNetV2-M + Disease Graph Attention

Report Date:	March 19, 2026	Report Time:	18:41:53
Model:	EfficientNetV2-M + DGA	Macro AUC:	0.8122 (95% CI: 0.8082–0.8161)
Image File:	test_xray.png	Findings:	5 positive finding(s)

FINDINGS SUMMARY

#	Finding	Probability	Margin	Confidence Tier
1	Pleural Thickening	93.1%	+13.1%	High Confidence
2	Effusion	90.0%	+10.0%	High Confidence
3	Cardiomegaly	93.6%	+8.6%	High Confidence
4	Fibrosis	90.9%	+5.9%	Borderline
5	Infiltration	64.5%	+4.5%	Borderline

1. CLINICAL INDICATION

The chest X-ray was performed as part of an AI-assisted screening protocol to evaluate for potential thoracic abnormalities.

2. TECHNIQUE

The examination consisted of a standard posterior-anterior (PA) chest radiograph with AI-assisted analysis using EfficientNetV2-M with Disease Graph Attention, which has a macro AUC of 0.8122.

3. FINDINGS

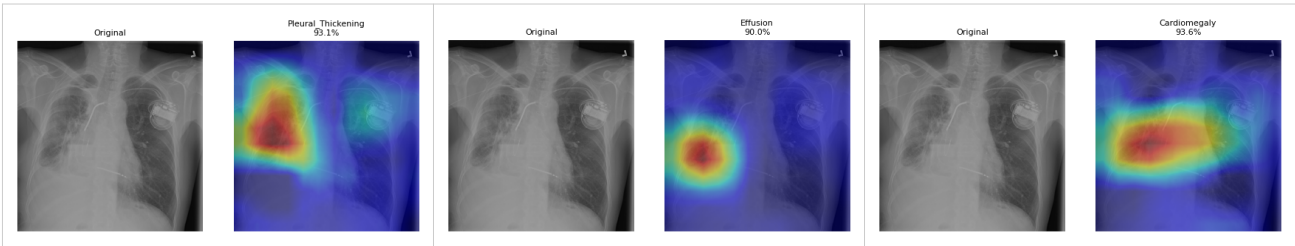
- 1. Pleural Thickening:** High Confidence (93.1% confidence, +13.1% margin). The AI model detected pleural thickening, primarily focusing on the left lung region (60.9% of activated area). This finding is consistent with an irregular density at the lung apex, characteristic of pleural thickening. The model's high confidence suggests a strong likelihood of this condition.
- 2. Cardiomegaly:** High Confidence (93.6% confidence, +8.6% margin). Cardiomegaly was detected, with the model focusing on the central region, including both lungs and the mediastinum. The cardiothoracic ratio is likely increased, suggesting cardiac enlargement. This finding is indicative of an enlarged cardiac silhouette, which may be associated with various cardiac conditions.
- 3. Effusion:** High Confidence (90.0% confidence, +10.0% margin). The AI model identified a pleural effusion, primarily in the left lung region (76.6% of activated area). This finding is consistent with blunting of the costophrenic angle, a characteristic sign of pleural effusion. The model's high confidence suggests a significant accumulation of fluid in the pleural space.

4. IMPRESSION

The most significant findings in this chest X-ray are the high confidence detections of pleural thickening, cardiomegaly, and effusion, which suggest a complex thoracic condition. The co-occurrence of these findings may indicate a chronic condition with cardiac and pleural involvement. The borderline findings of fibrosis and infiltration require clinical correlation to determine their significance. Therefore, it is recommended that these findings be correlated with clinical symptoms and additional diagnostic tests to confirm the diagnosis and guide management.

GRAD-CAM VISUAL ATTENTION MAPS (Top 3 Findings)

The following heatmaps illustrate the regions of the chest X-ray that most strongly influenced the AI model's predictions. Red/yellow regions indicate highest model attention.



COMPLETE PROBABILITY SCORES

Finding	Probability	Threshold	Status
Atelectasis	44.7%	80%	negative
Cardiomegaly	93.6%	85%	POSITIVE
Consolidation	48.6%	80%	negative
Edema	55.7%	85%	negative
Effusion	90.0%	80%	POSITIVE
Emphysema	8.3%	85%	negative
Fibrosis	90.9%	85%	POSITIVE
Hernia	6.3%	85%	negative
Infiltration	64.5%	60%	POSITIVE
Mass	60.0%	80%	negative
No Finding	12.3%	25%	negative
Nodule	66.1%	80%	negative
Pleural Thickening	93.1%	80%	POSITIVE
Pneumonia	56.7%	85%	negative
Pneumothorax	42.5%	80%	negative

5. DISCLAIMER

This report is generated by an automated AI system and must be reviewed and verified by a qualified radiologist before clinical use. The findings and impressions presented in this report should not be considered definitive until confirmed by a qualified healthcare professional.